



AMS HR System — Complete Build Manual

คู่มือสร้างระบบทีละขั้น สำหรับผู้เริ่มต้น / Step-by-step, for a beginner — no prior deployment experience assumed.

Covers the whole package: **Report** · **Org Chart (photos)** · **Employee Data (PDPA)** · **Leave & OT with 3-level approval + email reminders** · **Car Booking (Jira + LINE)** · **Stationery ERP (OFFICEMATE PO + reimbursement)** · **Users & Roles**

ANCA Manufacturing Solutions (Thailand) — HR · Version 2026-06-12 · Goes with the package `AMS_HRM_Complete.zip`

Honesty note: everything in this manual matches the code in the package. Where a step depends on YOUR accounts (Jira ids, SMTP password, Supabase URL) it says so explicitly — nothing is invented.

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\$0 — What you are building

One Streamlit web app (Python) with a database behind it, plus two small external helpers that you already built in earlier sessions: a **Google Apps Script hub** for car booking (talking to Jira and LINE) and your **OfficeMate catalogue data** feeding the ERP module.

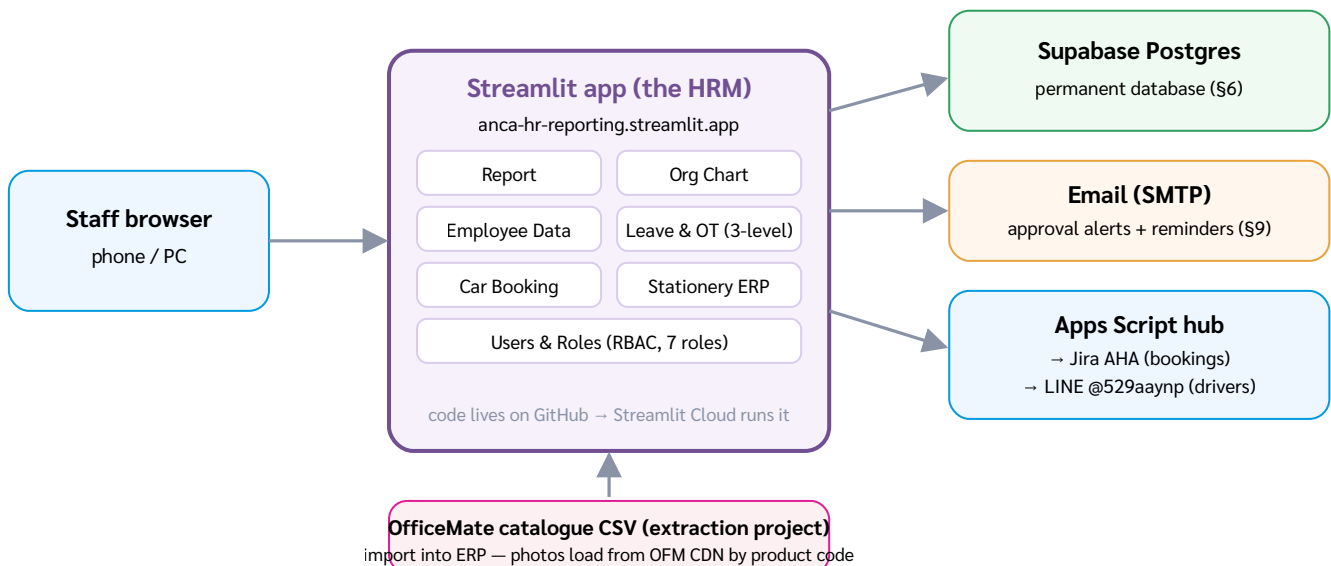


Figure 1 — the whole system. You only maintain ONE app; the green/orange/blue boxes are services it talks to.

Why this shape? Streamlit Cloud is free but its disk is wiped on every restart — that's why the database lives in Supabase (\$6). The car booking keeps its proven Jira/LINE backbone; the HRM is just its front door, so nothing you built before is thrown away.

§1 — Accounts & tools you need

What	Used for	Cost	Where
Python 3.11+	run the app on your PC	free	python.org
GitHub account	store the code; Streamlit Cloud reads from it	free	github.com
Streamlit Community Cloud	host the live app	free	share.streamlit.io (sign in with GitHub)
Supabase	permanent Postgres database	free tier OK; Pro \$25/mo for no-pause + daily backups	supabase.com
An email account for sending	approval notifications (\$9)	free (Microsoft 365 / Gmail App Password)	your IT
Jira (you have it)	car booking system of record — project AHA	existing	ancagroup.atlassian.net
LINE OA (you have it)	driver notifications	existing	@529aaynp · developers.line.biz

Security rule for the whole manual: passwords, API tokens and database URLs go ONLY into *Secrets* (Streamlit settings, Script Properties, GitHub repository secrets) — never into code files, never pasted into chat. If a token was ever shown in a chat, rotate it.

§2 — Run the app on your own computer first

Always make it work on your PC before touching the internet — fixing problems locally is 10x easier.

1 Unzip `AMS_HRM_Complete.zip` somewhere easy, e.g. `C:\ams-hrm`. You should see `app.py`, folders `lib/` `pages/` `assets/` `apps_script/` `scripts/` and `requirements.txt`.

2 Open **Command Prompt** (Windows key → type `cmd`) and go there:

```
cd C:\ams-hrm
```

3 Install the libraries (one time):

```
pip install -r requirements.txt
```

If `weasyprint` fails on Windows, that's OK — it's only for printing the government forms as PDF; everything else still runs. The print tab will offer HTML instead.

4 Start the app:

```
streamlit run app.py
```

A browser tab opens at `http://localhost:8501`.

5 Log in with the first-boot account `superadmin / ChangeMe!2026`. You will change this in §5.

Checkpoint [OK] you can see the home page and the left sidebar lists: System Admin, Report, Org Chart, Visitor Portal, Candidate Portal, Employee Data, My Profile, Leave & OT, Car Booking, Stationery ERP.

On your PC the app uses a temporary SQLite file (`data/hr.db`). That's fine for testing. The permanent database comes in S6 — the app shows a yellow warning until then, on purpose.

§3 — Put the code on GitHub

Why? Streamlit Cloud doesn't accept a zip — it watches a GitHub repository and redeploys automatically every time you change the code.

- 1 Sign in at github.com → top-right + → **New repository**. Name: `ams-hrm`. Visibility: **Private**. Create.
- 2 Easiest upload (no Git commands): in the new repo click **uploading an existing file**, drag in EVERYTHING from the unzipped folder (keep the folder structure), then **Commit changes**.
- 3 Check that `app.py` is at the repository ROOT (not inside an extra folder). If GitHub shows `ams-hrm/hr_app/app.py`, move files up one level.
- 4 **Do NOT upload** `.streamlit/secrets.toml` (real secrets) or `data/hr.db`. The package only contains `secrets.toml.example`, which is safe.

§4 — Deploy to Streamlit Cloud

- 1 Go to share.streamlit.io → sign in with GitHub → **New app**.
- 2 Repository: `your-name/ams-hrm` · Branch: `main` · Main file: `app.py`.
- 3 App URL: choose the name (yours is already `anca-hr-reporting`) → **Deploy**. First build takes a few minutes.
- 4 Open ⚙ → **Settings** → **Secrets** and paste the contents of `.streamlit/secrets.toml.example`, then fill the real values as you complete S6 (DATABASE_URL), S9 ([smtp]) and S10 ([car_booking]). Save — the app restarts itself.

Already have the app deployed? Your live `anca-hr-reporting` app keeps working. To upgrade it to this package, replace the files in its GitHub repo with these ones. One honest caveat: I rebuilt `pages/1 Report.py` as a clean baseline because I cannot open your deployed copy — if your live Report page has extra charts you added later, KEEP YOUR FILE and skip mine; nothing else depends on it.

§5 — First login & accounts

- 1 Log in as `superadmin / ChangeMe!2026`.

- 2 Open **System Admin** → **Users**. Create your own account: username, strong password, role `super_admin`, **Emp. No.** = **1021656**, your email. Then create a second `admin` account for daily HR work (admin cannot see salary — that separation is deliberate).
- 3 Create the `superadmin` replacement habit: after your own `super_admin` account works, change the bootstrap password — System Admin has the user list; set a long random password for `superadmin` and store it in your password manager (it stays as the emergency key).
- 4 **Link every user to an Emp. No.** (Users tab → "Link account ↔ employee"). This is what makes *My Profile*, the *3-level approval chain* and *email notifications* know who is who. Also fill each manager's **Email** here — approval mail goes to this address first.

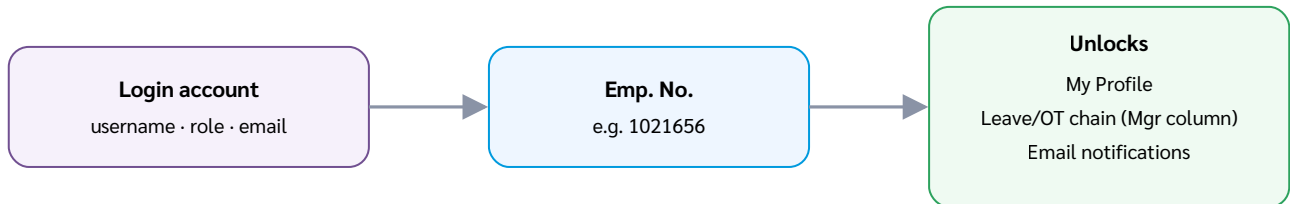


Figure 2 — the account → employee link is the key that makes the personal modules work.

S6 — Make data permanent: Supabase

This is the most important section. Without it, every redeploy/sleep of Streamlit Cloud wipes your database — the old "re-upload everything every 7 days" pain. With it, data lives in Supabase Postgres permanently.

- 1 supabase.com → New project. Region: **Southeast Asia (Singapore)**. Set a strong **database password** and save it.
- 2 In the project: **Connect** (top bar) → tab **ORMs/URI** → copy the **Transaction pooler** URI. It looks like:
`postgresql://postgres.abcdefgh:[PASSWORD]@aws-0-ap-southeast-1.pooler.supabase.com:6543/postgres`
- 3 Replace `[PASSWORD]` with your real password.
- 4 Streamlit Cloud → your app → Settings → **Secrets** → add one line at the top:

```
DATABASE_URL = "postgresql://postgres.abcdefgh:YourPassword@aws-0-ap-southeast-1.pooler.supabase.com:6543/postgres"
```
- 5 Save → app restarts → the yellow "temporary SQLite" warning on the home page disappears. The app creates all tables itself on first boot (you do NOT write SQL).
- 6 Re-create your users (\$5) and re-upload the MASTER (\$7) once — this is the one-time move from the throwaway database to the permanent one.

Free tier vs Pro: free pauses the database after ~1 week of no traffic (first visitor then waits ~1 minute while it wakes). Pro (\$25/mo) never pauses and adds daily backups — recommended once the company relies on it.

§7 — Load your data

7.1 Employee MASTER

- 1 **Employee Data → Bulk & Exports → Bulk upload.** Drop your `Employee List MASTER V.6` Excel directly — the parser recognises the MASTER layout (it mapped all 43 columns and imported 190 rows in testing) as well as the app's own flat template.
- 2 Review the preview, press **Import**. Salary columns are accepted **ONLY** if you are `super_admin` — for admin uploads they are ignored by design.
- 3 Spot-check 2–3 people in the Directory (we verified record 1021656 and 1021499 against the source during testing).

7.2 Photos & org chart

- 1 **Employee Data → Photos:** upload a face photo per employee — auto-cropped to 200×200 JPEG ≤100 KB.
- 2 **Org Chart** page then shows every photo automatically, and the chart toolbar has **Download JPG** for a 2x export of the current view.

7.3 OfficeMate catalogue → ERP

- 1 **Stationery ERP → Catalog admin → Import.** Upload the extraction master CSV/XLSX (the file produced by the catalogue-extraction sessions; columns = `product_code`, `item_no_model`, `category_type`, `description`, `brand`, `capacity`, `color`, `min_order_qty`, `price_thb`, `currency`, `image_file`, `source_catalogue`, `source_page`, `catalogue_note`).
- 2 Re-import after every extraction session — it upserts by `product_code` , so new rows add and existing rows update; nothing duplicates.
- 3 Retail product photos need no upload at all: the page builds the OfficeMate CDN URL from the code (`100-9755` → `.../products/large/1009755.jpg`).

§8 — Leave & OT: the 3-level approval

The approval route is **not configured by hand** — it is read live from the **Mgr column** of the employee master (the same data as the org chart): L1 = direct manager, L2 = the manager's manager, L3 = one more level up.



Reject at ANY level → request becomes rejected, remaining levels are skipped, the staff is emailed with the approver's note

Shorter chains end early (e.g. a GM direct-report has only L1) · nobody can approve their own request

Figure 3 — the 3-level flow, exactly as verified in testing against the real MASTER.

Daily use

Who	Where	What they see
Staff	Leave & OT → Submit	their own chain shown above the form; after sending, "My requests" shows the live trail L1 [approved] → L2 [waiting] → L3 [-]
Manager	Leave & OT → My approvals	only items at THEIR level, with Approve/Reject + note; approving forwards it (and the email) to the next level
HR	Leave & OT → HR admin	the whole pipeline, who's sitting on what, and the reminder button (§9)

For the chain to resolve, three things must be true: (1) the staff's login is linked to their Emp. No. (§5), (2) the Mgr column in the master holds the manager's name as written in the master (titles like Mr./Ms. are ignored automatically), (3) the manager exists as an active employee. If no manager is found the request auto-approves and is flagged in HR admin — visible, never stuck.

§9 — Email notifications + automatic reminders

9.1 One-time SMTP setup

1 Get a sending mailbox from IT — e.g. `hr-noreply@...` on Microsoft 365 (host `smtp.office365.com`, port 587) or a Gmail with an **App Password** (host `smtp.gmail.com`, port 587).

2 Streamlit Cloud → Settings → Secrets → add:

```
[smtp]
host = "smtp.office365.com"
port = 587
user = "hr-noreply@yourdomain.com"
password = "the-app-password"
from = "AMS HR System <hr-noreply@yourdomain.com>"
app_url = "https://anca-hr-reporting.streamlit.app"
```

3 That's it. From now on: new request → email to L1; each approval → email to the next level; final decision → email to the staff. If SMTP is missing the app still works and simply says "email skipped" — it never crashes on mail problems.

9.2 Who gets the mail

The approver's address is looked up in this order: the **Email on their user account** (§5) first, then the employee master's email field. The HR admin pipeline table shows "(!) none" next to any approver with no address — fix those in System Admin.

9.3 Reminders — button and automatic

1 **Manual:** Leave & OT → HR admin → set the age threshold (default 24 h) → **Send reminders now**. Each manager gets ONE digest listing all their waiting items — not one mail per item.

2 **Automatic daily (recommended):** in your GitHub repo create the folder/file `.github/workflows/reminders.yml` and paste the contents of `.github_workflow_reminders.yml` from the package (it runs `scripts/send_reminders.py` every day 09:00 Bangkok).

3 In GitHub: repo → Settings → Secrets and variables → Actions → add `DATABASE_URL`, `SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASSWORD`, `SMTP_FROM` (same values as the app secrets).

4

Test it once: repo → Actions → "Daily approval reminders" → **Run workflow**. The log lists every reminder sent. An item is re-reminded at most once per day.

§10 — Car Booking (Jira + Apps Script + LINE)

The booking page now lives INSIDE the HRM (Car Booking in the sidebar) — but the engine is the proven external stack you already built. This section wires them together. If your Apps Script hub from the earlier sessions is already deployed, you only need steps 6–7.

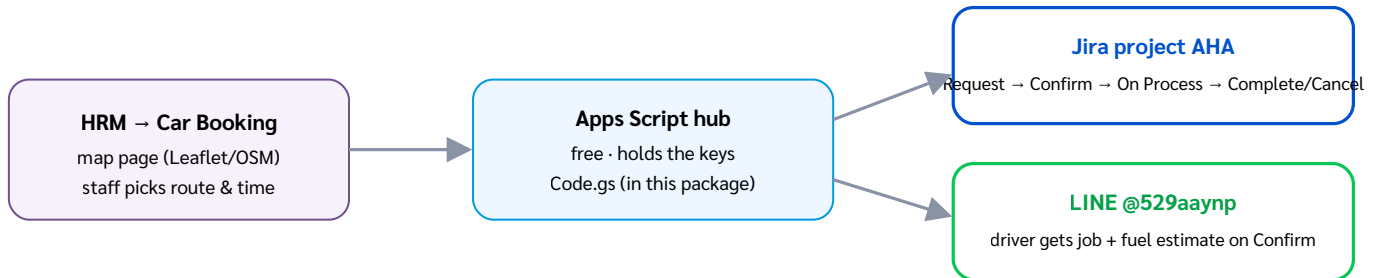


Figure 4 — booking flow. Google Sheet (settings/cars/drivers tabs) feeds fuel price, km/L and driver LINE ids — all editable without touching code.

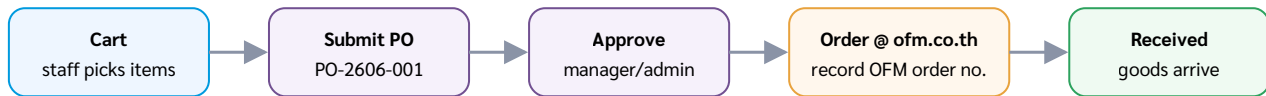
- 1 Jira:** you need space-admin on AHA. Work type **Car Booking** with your 22 custom fields exists already (customfield_15673–15694), and the workflow uses your corrected status names: **Request** → **Confirm** → **On Process** → **Complete / Cancel**.
- 2 Tokens:** Jira API token (id.atlassian.com → Security → Create API token) and the LINE **channel access token** for @529aaynp (developers.line.biz). If the old ones were ever pasted into a chat, create fresh ones now.
- 3 Google Sheet:** tabs `settings` (row: `fuel_price_thb | 35`), `cars` (A plate · B model · C seats · D active · E km/L), `drivers` (A name · B LINE userId). Copy the Sheet ID from the URL.
- 4 Apps Script:** script.google.com → New project → paste `apps_script/Code.gs` from this package → Project Settings → **Script properties:** `JIRA_EMAIL`, `JIRA_TOKEN`, `LINE_TOKEN`, `SHEET_ID`.

Fill the F map. The file ships with `customfield_XXXXX` placeholders because I can't read your live Jira from here — your deployed copy from the earlier session already has them filled, so copy that F block over, or re-list ids at `https://ancagroup.atlassian.net/rest/api/3/field`. Keep your three fixes: ONE combined "Pickup/Dest lat,lng" field; Purpose & Priority as plain text; Start/End via the `isoDT()/hhmm()` datetime helpers.
- 5 Deploy:** Deploy → New deployment → **Web app** → Execute as **Me** → Who has access **Anyone** → copy the `/exec` URL. Then Jira Automation: when status → Confirm, send web request (GET) to `<exec URL>?action=notify&key={{issue.key}}`. Optional: add a daily 07:00 time trigger on `dailySummary` for the drivers' morning schedule.
- 6 Connect the HRM:** Streamlit Secrets →

```
[car_booking]
apps_script_url = "https://script.google.com/macros/s/.../exec"
jira_board_url = "https://ancagroup.atlassian.net/jira/core/projects/AHA/board"
```
- 7 Test:** open Car Booking in the HRM → book a trip to "U-Tapao" → a Jira issue appears in **Request** → move it to **Confirm** with car+driver filled → the driver's LINE receives the job with the fuel estimate (distance ÷ that car's km/L × the Sheet's fuel price).

§11 — Stationery ERP

A · Purchase order (company buys from OFFICEMATE)



B · Reimbursement (staff already paid with their own money)



Figure 5 — the two ERP flows, both fully audited. Tabs in the module map 1-to-1 to these boxes.

Tab	Who (capability)	Does
Catalog	everyone (erp.browse)	search by code/description/brand/category, real OFM photo, add to cart
Cart & submit	staff (erp.request / erp.reimburse)	review cart, free-text lines for uncatalogued items, submit PO; submit reimbursement with receipt photo/PDF
My POs & claims	staff	own history with statuses
Approvals	manager/admin (erp.approve)	approve/reject POs and claims, see the receipt image inline
Purchasing	admin (erp.purchase)	place the real order on ofm.co.th, record the OFM order no., mark received
Finance	finance (erp.pay)	pay approved claims, record transfer reference
Catalog admin	admin (erp.manage_catalog)	import/refresh the extraction CSV (§7.3)

§12 — Roles & permissions reference

Role	Meant for	Key powers (defaults — Super Admin can reshape)
visitor	guests, applicants	Visitor Portal, job application, car booking
viewer	every staff member	Report, Org Chart, My Profile, submit leave/OT, car booking, ERP browse/request/reimburse
supervisor	line leaders	viewer + approve leave/OT at their level
manager	department managers	supervisor + approve ERP POs/claims
finance	accounting	viewer + pay reimbursements
admin	HR team	full employee data (NO salary), candidate management, bulk upload, exports, leave admin + reminders, car admin, ERP purchasing & catalog, users
super_admin	you	everything, including salary view/edit

Deliberate choices: salary is super_admin-only (admin cannot see it, even in bulk upload — salary columns in an admin's file are ignored); PDPA-sensitive fields never appear in the External export; every change is in the audit log (System Admin).

\$13 — Maintenance & troubleshooting

Symptom	Cause	Fix
Yellow "temporary SQLite" warning	DATABASE_URL not set	\$6
Data vanished after redeploy	was still on SQLite	\$6, then re-upload MASTER once
"Your login is not linked to an Emp. No."	account ↔ employee link missing	System Admin → Users → Link (\$5)
Approval chain empty / wrong	Mgr name doesn't match an active employee	fix the Mgr cell in Employee Data (or re-upload MASTER)
"email skipped — SMTP not configured"	[smtp] secrets missing	\$9.1
Approver shows "(!) none" in HR admin	no email on the user account	System Admin → Users → Email
Reminders not arriving daily	workflow file or GitHub secrets missing	\$9.3; check repo → Actions logs
Car booking page says hub not configured	[car_booking] secret missing	\$10 step 6
Booking fails "Jira 400"	F map ids wrong/placeholder	\$10 step 4
Driver got no LINE message	userId blank in drivers tab, or automation rule missing	\$10 steps 3 & 5
ERP photo blank	item is premium/made-to-order (no retail CDN photo)	expected — catalogue note says "Contact admin for customization"
App very slow first visit of the day	Supabase free tier waking up	expected on free; Pro removes it

Routine

- 1 Weekly:** HR admin pipeline — chase anything pending > 3 days (the reminder cron already nags daily).
- 2 After each catalogue-extraction session:** re-import the master CSV (\$7.3).
- 3 Monthly:** Supabase dashboard → Database → check size; download a backup (Pro does this daily automatically).
- 4 When someone joins/leaves:** Employee Data first (so chains and org chart stay true), then their user account in System Admin.
- 5 Never** edit secrets into the code. Rotate any token that ever appeared in a chat or screenshot.

End of manual · AMS HRM 2026-06-12 · Built and tested with: MASTER V.6 import (190 rows), 3-level chain verified on real reporting lines, PO/reimbursement lifecycles, RBAC matrix, app boot. The Code.gs F map and your live Report page are the two items only you can finish — both flagged honestly above (\$4, \$10).